

(ii) 1. The oscillation now experiences greater damping hence amplitude will decrease with time.

2. Equilibrium position is at a new position when length of spring is 0.275 m instead of 0.30 m.

2 (a) (i) Power of heater $= \frac{V^2}{R} = \frac{12^2}{2.4} = 60 \text{ W}$

(ii) Heat supplied by heater = Heat gained by water

$$0.14 \times c \times 20 = 60 \times 210$$

$$c = 4500 \text{ Jkg}^{-1}\text{K}^{-1}$$

(b) Heat loss to the surroundings.

3 (a) (i) From 0 to 0.1 V, zero current flowing through. The diode needs to be forward